Orthopedic Infections: A Clinical Pharmacy Perspective

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Definition of Osteomyelitis

- **Definition**: Inflammation of bone and bone marrow caused by an infecting organism.
- **Classification**:
  - Acute (<2 weeks)
  - Subacute (2-6 weeks)
  - Chronic (>6 weeks)

Acute Osteomyelitis – Key Organisms

- **Common Pathogens**:
  - **Staphylococcus aureus** (MSSA, MRSA) >>> **MC 90%**
  - **Others:** Streptococcus species, Gram-negative organisms (E. coli, Pseudomonas)
  - **Unusual:** Salmonella (in sickle cell anemia patients), pseudomonas in IV addicts and wounds

Acute Osteomyelitis – Risk Factors

- **Endogenous**: Diabetes, sickle cell disease, malnutrition, immunosuppression, obesity, smoking, systemic infections
- **Exogenous**: Trauma, surgery, foreign implants.

Acute Osteomyelitis – Sites

- Children:
  - Metaphysis (mainly around knee)
- Adults:
  - Spine

Acute Osteomyelitis – Route of infection

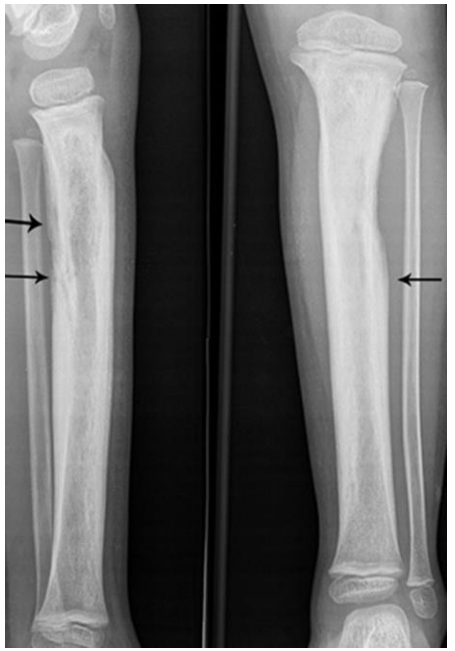
- I. Hematogenous spread
- II. Local spread
- III. Direct inoculation

Acute Osteomyelitis – Clinical Presentation

- **Symptoms**: Fever, localized pain, swelling, tenderness.
- **Lab Findings**:
  - ↑ WBC, ESR, CRP
  - Blood cultures (positive in 50% of cases).







Acute Osteomyelitis – TTT

1- General supportive treatment:

- Bed rest, Fluids.
- Analgesics, Anti-inflammatory.

2- Splintage:

- Skin traction or slabs
- Helps to ↓ pain, contractures

3- AB

4- Operative treatment

1. Failure to respond to medical treatment
2. Subperostieal abscess
3. Pus on aspiration
4. Chronic infections

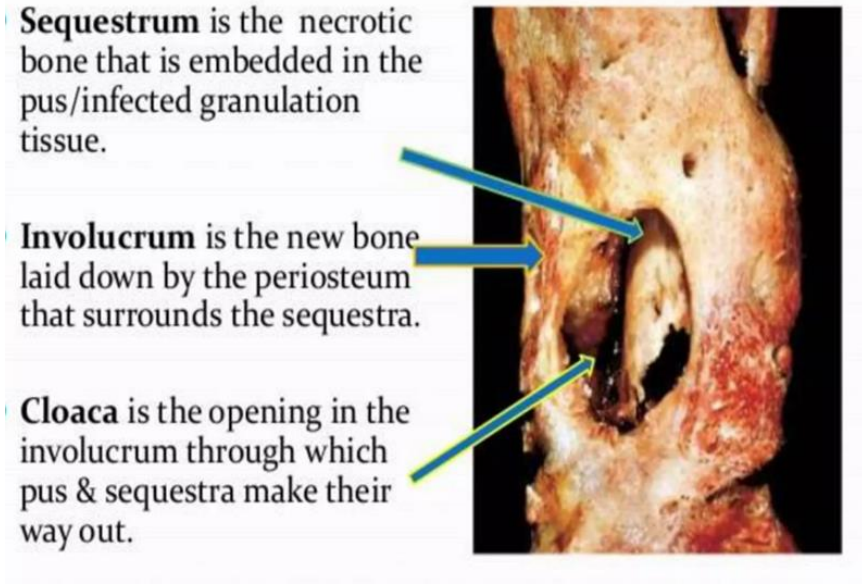
Acute Osteomyelitis – Antibiotic Therapy

- **Empirical Therapy**:
  - **MSSA**: Nafcillin, cefazolin, clindamycin.
  - **MRSA**: Vancomycin, linezolid.
  - **Gram-negative**: 3rd-gen cephalosporins (ceftriaxone), fluoroquinolones.
- **Duration**: 4-6 weeks (IV + oral).

Chronic Osteomyelitis – Treatment

- **Key Principles**:
  - Usually occurs as a sequel to acute hematogenous, postoperative or post-traumatic OM.
  - Cavities containing dead bone spicules (**Sequestra**) that may act as a media for bacterial adhesion
  - **Surgical debridement + antibiotic therapy.**
  - **Antibiotics**: Tailored to culture results.
- **Duration**: 4-6 weeks.





Septic Arthritis – Overview

- **Definition**: Purulent inflammation of synovial membrane within a joint.
- **Common Pathogens**:
  - **Staphylococcus aureus** (most common).
  - **Streptococcus species**, **Gram-negative organisms**.
- **Treatment**:
  - **Surgical emergency!!!!!!!**

Pyogenic Vertebral Osteomyelitis (PVO)

- **Common Pathogens**:
  - **Staphylococcus aureus** (50%).
  - **Gram-negative organisms** (E. coli, Pseudomonas).
- **Treatment**:
  - **Empirical Therapy**: Vancomycin + ceftriaxone.
  - **Duration**: 6 weeks.

Tuberculous Osteomyelitis – Pott's disease

- **Pathogen:**
  - **Mycobacterium tuberculosis** (acid-fast bacillus).
- **Pathology**:
  - Granuloma formation with caseation necrosis.
  - Cold abscess formation (may drain via sinuses)

Tuberculous Osteomyelitis - Clinical Presentation:

- **Systemic Symptoms**: Night sweats, fever, weight loss, malaise.
- **Local Symptoms**:
  - Back pain (most common symptom).
  - Spinal deformity (kyphosis, gibbus deformity).
  - Neurological deficits (if spinal cord compression occurs).
- **Investigations:**
  - **Lab Tests**:
    - **Tuberculin Skin Test (TST)**: Positive in active or past infection.
    - **Interferon-Gamma Release Assays (IGRAs)**: More specific than TST.
  - **Imaging**:
    - **X-ray**: Vertebral destruction, disc space narrowing, kyphosis.
    - **MRI**: Paravertebral abscess, spinal cord compression.
    - **CT-guided biopsy**: Confirms diagnosis (acid-fast bacilli on smear or culture).

Tuberculous Osteomyelitis - TTT

- **First-line Drugs**:
  - **Isoniazid (INH)**: 5 mg/kg/day (max 300 mg).
  - **Rifampin**: 10 mg/kg/day (max 600 mg).
  - **Pyrazinamide**: 15-30 mg/kg/day (max 2 g).
  - **Ethambutol**: 15-25 mg/kg/day.
- **Duration**:
  - **Intensive Phase**: 2 months (INH + rifampin + pyrazinamide + ethambutol).
  - **Continuation Phase**: 4-7 months (INH + rifampin).
- **Monitoring**:
  - Liver function tests (LFTs) for hepatotoxicity (INH, rifampin, pyrazinamide).
  - Visual acuity for ethambutol-induced optic neuritis.